Model preparation
Supporting Document: `IndigoCarbon_US-1_2023_0073`

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1 Model Version

Per the SEP Model Requirements and Guidance v1.1a (herein "model requirements"), the model version is "A uniquely traceable record of source code, internal parameters, and ancillary inputs that are needed to reproduce a given model output (collectively the "model files"). A model version must change any time there is a change in any of the model files."

Per the model requirements, when submitting a validation report it must be shown that all requirements for a specific project have been met, including proof that the same model version and parameter sets are used, and that all project domain and crop functional group / practice category combinations have met minimum requirements for model validation. (See `IndigoCarbon_US-1_2023_0046` for the specific crop functional group / practice category combinations found in project CAR1459).

The model version validated for use in CAR1459 is DayCent-CR Version 1.1.0. The validation report for DayCent-CR Version 1.1.0 (`IndigoCarbon_US-1_2023_0046`) has passed the independent review process, per Section 1 of the model requirements. DayCent-CR Version 1.1.0 was used in crediting simulations for CAR1459. As proof of this, a verifier may compare the model files submitted to the third-party entity (detailed below) to the model files submitted with the monitoring plan, i.e. associated with the approved validation report (`IndigoCarbon_US-1_2023_0046`).

The model files for DayCent-CR Version 1.1.0 consist of two components. Each component is version-controlled independently, and together represent the validated DayCent-CR Version 1.1.0:

1. Version 1.1.0 of the DayCent-CR model executable, compiled from revision 32aa16b of Indigo Ag's DayCent code repository. (see Model Guidance Section 5).
2. Version 3.0 of the DayCent-CR model parameters, with fixed components from revision a7f2701 of Indigo Ag's DayCent validation data repository and probability distributions of calibrated components from revision 8326da36d5of Indigo Ag's cross-validation pipeline.

When executing model simulations for project CAR1459, Indigo submits input data to the model using the DayCent-CR API Version 2.0.0 (`IndigoCarbon_US-1_2023_0049` *Model Provider Documentation*). This API is set up in such a way to ensure model inputs are provided as required by the model, and simulations are run using the model executable and parameters described in the two bullets above. However, the API was not needed to run calibration or validation simulations and is thus not included in the DayCent-CR Version 1.1.0 validation report (`IndigoCarbon_US-1_2023_0046`). The validation report should be applicable to any result obtained from DayCent-CR Version 1.1.0 whether it is run directly, as it was for calibration and validation simulations, or accessed through any technically compatible version of the DayCent-CR API.

Note that model predictions for any Crop Functional Group / Practice Category combination simulated for credits must not exceed 5,000 g m^{-2}, per the interpretation of the validation results reported in IndigoCarbon_US-1_2023_0046 (see section entitled *Restrictions on application of model*). The enforcement of this restriction is described in `IndigoCarbon_US-1_2023_0075`.

2 Model Spin-up

In accordance with best practice in biogeochemical modeling, DayCent-CR Version 1.1.0 requires the model to be spun up over a period of thousands of years prior to the simulation of baseline and project scenarios. This allows modeled pools of carbon to come to an approximate equilibrium in preparation for simulating the effects of contrasting scenarios of management. The spin-up period for DayCent-CR Version 1.1.0 is performed under the direction of model developers at Soil Metrics LLC, and is the same procedure used in calibrating and validating the model `IndigoCarbon_US-1_2023_0046`).

The spin-up procedure used to prepare all baseline and project scenario simulations at a given site includes the following steps:

1. The model is initialized with site latitude and longitude, and soil physical and chemical properties provided by Indigo for the site location (`IndigoCarbon_US-1_2023_0030`) with the exception of soil carbon pools which are initialized based on literature-derived results (Burke 1989).
2. Native grassland is grown for a period of several thousand years until the model reaches a steady-state within the soil carbon pools, consistent with the US National GHG Inventory and current implementation of COMET-Farm (USEPA 2021).
3. Once equilibrium is achieved, a "plow-in" period is simulated through a series of historically reconstructed model simulations based on land uses and agronomic practices that were typical for the region in the following time periods. This approach has been leveraged in previous research using DayCent versions similar to DayCent-CR Version 1.1.0 (Parton et al. 1998, Parton et al. 2005, Hartman et al. 2011, Gurung et al. 2021, US-EPA 2021).
    1. European settlement to 1950: This period approximates agricultural practices before agricultural mechanization, improved crops, and the proliferation of chemical fertilizers during the green revolution.
    2. 1951-1980: Simulates the rapid improvement in crop varieties and proliferation of chemical fertilizers during the green revolution (1951-1979).
    3. 1981-2000: Simulates the period after the green revolution when modern hybrid crops and high levels of chemical fertilizers became common.
4. Beginning in calendar year 2000, full management information is required from Indigo including crop plantings and harvests, tillage, and nutrient applications (see `IndigoCarbon_US-1_2023_0030`).
5. The spin-up period continues until the start of the baseline and/or project simulation, at which point the model will be re-initialized using the SOC measurement (`IndigoCarbon_US-1_2023_0001`), and the parameter set switches from the spin-up parameter set to the validated parameter set (as described in *Parameter Sets* below). At this point the baseline or project scenario simulation commences.

No management data are required from the project developer to support Steps 1–3 of the model spin-up period. Files used to perform the model spin-up are compiled and maintained using best available data (including SME values). For the period between 2000 and the start of the historical baseline period, the data from the historical baseline period are cycled through repeatedly until the start of the actual historical baseline period. From there, the historical baseline data are provided corresponding to their actual calendar years (see example in Figure 1). Safeguards were

added to the code used to generate data for the period between 2000 and the start of the historical baseline period to ensure that events occurring after the current calendar year's cultivation cycle end are fully represented.

Meteorological data used throughout the spin-up period come from the North American Land Data Assimilation System (NLDAS) (`IndigoCarbon_US-1_2023_0069, IndigoCarbon_US-1_2023_0030`). Prior to 1975, the NLDAS data are cycled through repeatedly in sequence. NLDAS data are available beginning in 1979, and thus the years 1979-1982 are used to simulate years 1975–1978. Beginning in 1979, the actual sequence of NLDAS data is used, corresponding with the true calendar year sequence of the remainder of the simulation.

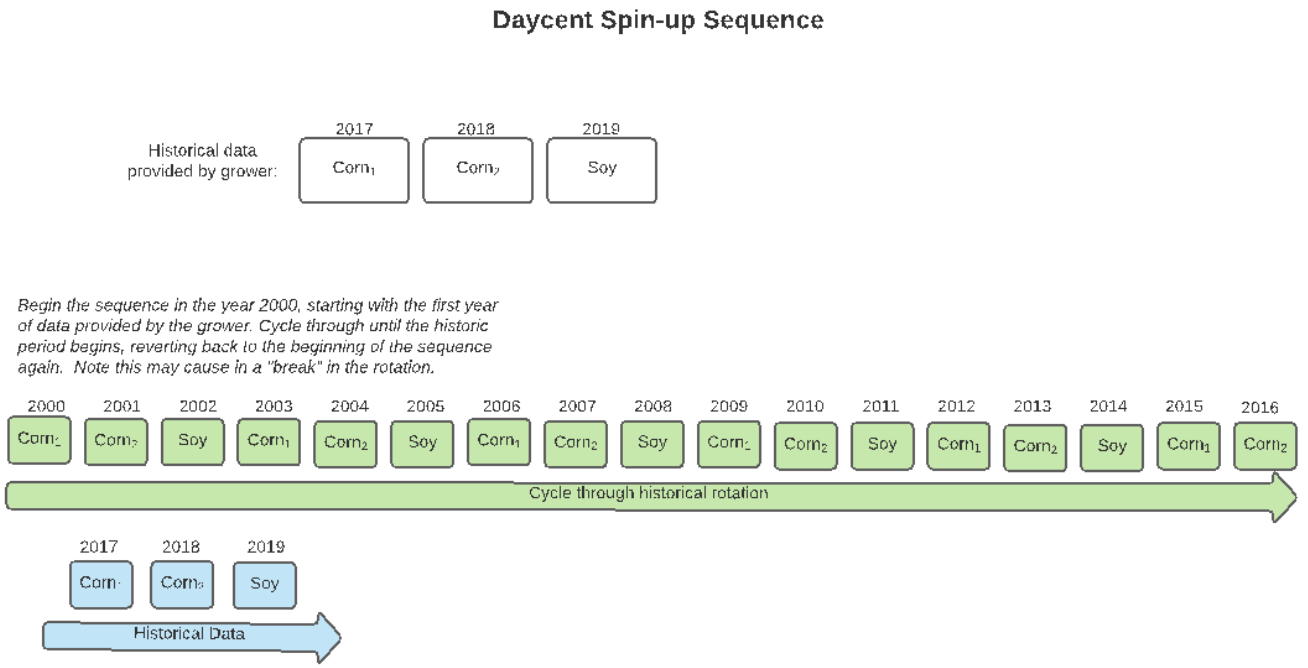


Figure 1: Example of a spin-up sequence of crops using a hypothetical rotation.

3 Parameter Sets

Per the Model Requirements and Guidance v1.0a, a parameter set is defined as "the set of mathematical values and constants contained in a model that characterizes the biophysical and biogeochemical system being represented. Depending on the model and the application, the value of an individual parameter within a given parameter set might be defined as a single number, a set of numbers with declared rules for choosing from them, and/or a probability distribution."

There are two internal parameter sets used in crediting simulations. The first set is used for the spin-up period (see *Spin Up* above) up to the year of the SOC measurement. This parameter set was originally derived for with the version of DayCent branched to create DayCent-CR, which was used in the U.S. National Greenhouse Gas Inventory (USEPA 2021) but contains an expanded crop and practice list developed for the Comet-Farm tool (https://comet-farm.com). At the point of re-initialization using the SOC measurement, the parameter set switches to the parameter set with outputs from the cross-validation procedure, as declared and documented in detail in Appendix E of the validation report (`IndigoCarbon_US-1_2023_0046`) and provided in file CAR1459_model_val_DayCentCR_1.0_appendices.pdf. This second parameter set is used for baseline and project crediting simulations within the Soil Enrichment Protocol.

4 Backmodeling

4.1 Summary

The project requires modeled soil organic carbon (SOC) pools to be initialized (see `IndigoCarbon_US-1_2023_0030`-*Model Input Mapping; Section 10- Soil Inputs*) with a measured SOC value at the start date of the cultivation cycle when qualifying additional practices were first adopted. As described in `IndigoCarbon_US-1_2023_0048`, cultivation cycles are modeled at the level of the Management Zone in which the cultivation cycle is completed. Per SEP Section 5, there may be discrepancy between the start date and when soil samples were taken. In this project, SOC has been measured after the start date (in some cases >1 year after the start date). DayCent-CR Version 1.1.0 model structure only allows initialization with measured SOC at the start of a new simulation or at the start of an extension from a previously completed simulation (i.e., the extension into the with-project period of simulation from equilibrium and historical baseline spin up simulations). Initialization therefore always falls on January 1st of the year in which the start date occurs (see Figure **??** `IndigoCarbon_US-1_2023_0048` for a diagram of how DayCent-CR Version 1.1.0 initial with-project and baseline simulations are constructed). This discrepancy in dates is not a problem, if it does not lead to a positive carbon generation and significantly increase the effect size. The remainder of this section describes how we tested for these concerns with the discrepancy and showed that no correction was needed.

One way to assess the impact of the discrepancy between dates is to compare model performance when using the SOC measurement to initialize the model directly, regardless of when it was taken, versus using an initialization value determined by backmodeling of SOC from the point in time the SOC measurement was taken back to the start date. DayCent-CR Version 1.1.0 can only run forward in time from when it is initialized, so backmodeling requires identifying an initial SOC value such that modeled and measured SOC closely match on the date SOC was sampled. For this assessment we used a brute force method to complete backmodeling, wherein multiple candidate model outputs were evaluated for a single soil sample point, to determine a single best initial SOC stock that provided the minimum absolute differences between the modeled and the measured SOC stocks on the date at which the soil was sampled. The model results from each simulation (i.e., the simulation using the soil sample measurement directly to initialize the model, versus the simulation using the adjusted soil sample measurement generated via backmodeling) were then compared based on estimated effect sizes generated from the same post-processing steps used in the Carbon Data Pipeline (`IndigoCarbon_US-1_2023_0065`). This comparison was evaluated to determine the sensitivity of carbon credit generation to the different initial SOC values.

We showed that the differences were negligible between SOC initialization approaches (Figure 2), and the 95% confidence intervals (i.e., 2.5 and 97.5 quantiles) of differences fall between -0.0032 and 0.0155 tCO2e/acre for cover-crop, -0.0050 and 0.0041 tCO2e/acre for no-till, and -0.0069 and 0.0051 for both no-till and cover-crop adaptation. These differences in effect size are small enough that we determined that backmodeling adjustments of initial measured SOC are unnecessary for modeling SOC using DayCent-CR Version 1.1.0, as it would not have a material impact on credit quantification.

Further, the brute force method for backmodeling is computationally intensive, would add complexity to the modeling process, and has no material impact on credit quantification. Therefore, backmodeling was not used to support crediting in CAR1459. Instead, measured soil samples of SOC were used at model initialization using the same values originally obtained in the laboratory.

4.2 Model Simulations and Analysis

The analysis was conducted using DayCent-CR 1.0.2 and Soil Metrics API DCR1.0.2 build 1.0.1.7. A sample of field locations and a single soil sampled point per field were selected that cover 12 LRRs and three regenerative practice changes (adoption of no-till, cover-crop and both no-till + cover-crop) and consists of a total of 106 unique field locations. The sample size and spatial distribution are provided in Table **??** and Figure 2, respectively.

Table 1: Total number of sampled point locations in each LRR by practice category.

LRR	cover-crop	no-till	no-till + cover-crop	Total
F	4	9	3	16
HN	3	10	2	15

Table 1: Total number of sampled point locations in each LRR by practice category.

LRR	cover-crop	no-till	no-till + cover-crop	Total
HS	3	6	2	11
IN	-	1	1	2
J	1	-	-	1
K	-	2	2	4
L	3	2	2	7
M	10	10	10	30
N	2	1	1	4
O	3	1	1	5
PW	2	1	1	4
R	-	3	3	6
TE	-	-	1	1
Total	32	45	30	106

The brute force backmodeling simulation is performed as follows:

Steps:

1. Run the model for the project scenario without the Monte Carlo option, and initialize with measured SOC. This option without Monte Carlo simulates the project scenario with the posterior mode and is considered the single best estimate of model output.
2. Calculate the Rate of Change of modeled SOC between the initialized (i.e., measured) and modeled SOC at the end of the model simulation as follows:

$$RoC = \frac{(modSOC - initSOC)}{initSOC}$$

where, RoC, is the rate of change in SOC from the beginning of the simulation to the end of the simulation for the project scenario, $modSOC$ is the modeled SOC stocks (g C m^{-2}) at the end of the simulation, and $initSOC$ is the measured SOC stocks (g C m^{-2}) used for model initialization.

3. Determine the new initial SOC stock for the analysis; this is the midpoint for initial candidates and calculated following:

$$midPoint = \frac{initSOC}{1 + RoC}$$

4. Determine an upper and a lower bound around the $midPoint$. These upper and lower bounds are used to draw candidates for SOC initialization using the following rules:
    a. If the absolute rate of change ($|RoC|$) is <0.025, set the lower and upper bounds to 0.95 and 1.05, respectively.
    b. If the absolute rate of change ($|RoC|$) is >=0.025, set both lower and upper bounds to twice the $|RoC|$. For example, for $|RoC|$ value of 0.05, the lower and upper bounds would be 0.9 and 1.1, respectively.
    c. Round the lower and upper bounds to two decimal places and take a gridded sample with 0.005 increments. The grided sample will provide at least 9 candidate samples and the increases with the rate of change
5. Draw a list of candidates for initial SOC values by multiplying the gridded sample from Step 4(b) and the $midPoint$ from Step 3.

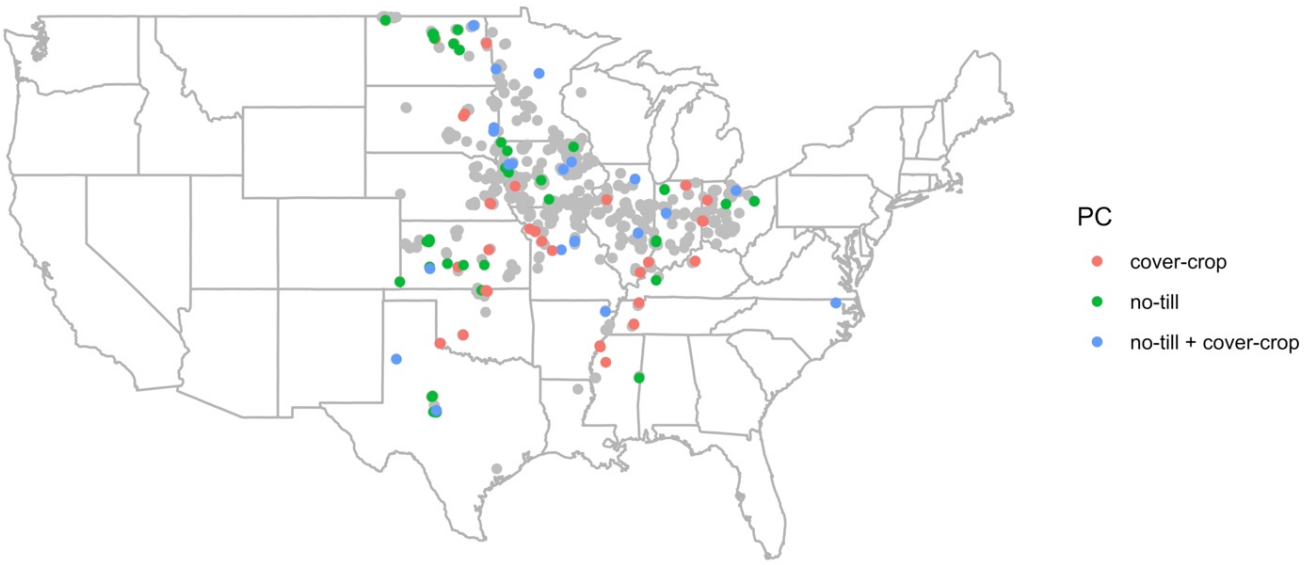


Figure 2: Representative point locations used for the analysis of **backmodeling** approach to adjust initial SOC measurements. Grey dots are soil sampled fields in the first issuance but not selected for this analysis, shown for context. Color dots are field locations used for the analysis with adaptation of regenerative practice change with (red) Cover-Crop, (green) No-Till, and (blue) both No-Till and Cover-Crop.

6. Run DayCent-CR Version 1.1.0 using each candidate SOC value, using the API (similar to Step 1, but now with Monte Carlo options turned on) for both project and all baseline scenarios.
7. Calculate differences between modeled and measured SOC stock at measurement date for all initial SOC candidates and determine the best candidate that yields the minimum absolute differences. If the measurement is taken after the last day of model simulation (no fields are simulated after 2020-12-31 and some fields have soil sampled taken during 2021) compare modeled and measured on the last day model output is available.
8. Estimate Effect Size using post-processing of model outputs and credit calculation as outlined in `IndigoCarbon_US-1_2023_0065` and calculate the difference in effect size between two initialized SOC values; (1) measured SOC and (2) best candidate during backmodeling.

4.3 Results and Conclusion

The brute force method was successful in estimating initial SOC values, providing a modeled value within +/-0.327 % of the measured SOC for all sampled fields. We also analyzed the effect of initialization using the measured SOC instead of backmodeled SOC value by taking the effect size differences (Best Candidate – Measured SOC) using the two initialization methods. The analysis suggest that the effect sizes have a low sensitivity to initial SOC, with effect size differences varying between -0.0032 and 0.0155 tCO2e/acre for cover-crop, -0.0050 and 0.0041 tCO2e/acre for no-till, and -0.0069 and 0.0051 tCO2e/acre for both no-till and cover-crop adaptation (Figure 3). These differences are small enough to consider backmodeling adjustments to SOC measurements unnecessary to initialize model simulations of SOC, as it would not have a material impact on credit generation. For the sake of simplifying modeling processes, backmodeling was not used to adjust measurements of SOC to initialize the model, instead the model was initialized directly with measured SOC.

5 References

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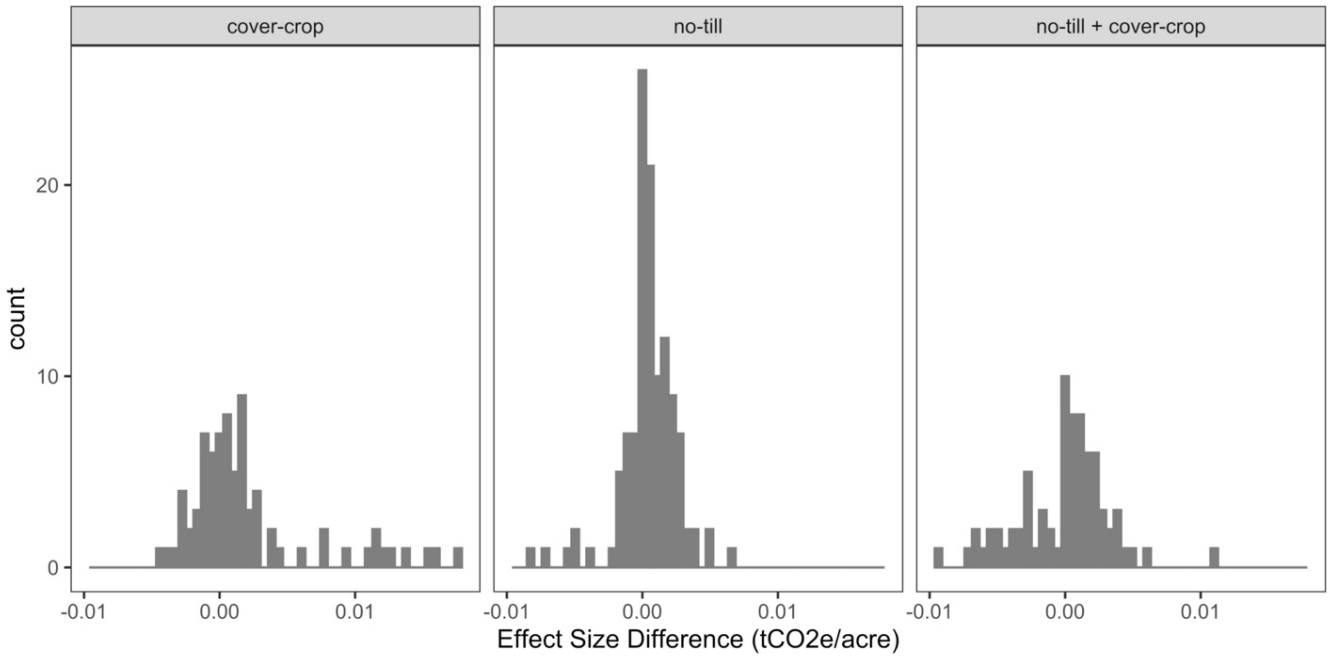


Figure 3: Difference in Effect Sizes between the model simulation initialized with measured SOC and initial SOC determined by brute force method for backmodeling.

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6 Summary of Changes

Section	Date	Summary of Changes
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